

Scheduling Algorithms

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1. FCFS Scheduling algorithm

```
rohitsai@MSI: ~/OS
GNU nano 8.7.1
#include <stdio.h>

int main() {
    int n, i;
    int bt[20], wt[20], tat[20];

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time for each process:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }
    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}
```

Output:

```
rohitsai@MSI: ~/OS
rohitsai@MSI:~$ cd OS
rohitsai@MSI:~/OS$ nano FCFS.c
rohitsai@MSI:~/OS$ gcc FCFS.c -o FCFS
rohitsai@MSI:~/OS$ ./FCFS
Enter number of processes: 3
Enter burst time for each process:
P1: 3
P2: 5
P3: 2

Process BT      WT      TAT
P1      3      0      3
P2      5      3      8
P3      2      8     10
```

2. SJF_NONpreemptive

```
GNU nano 8.7.1
#include <stdio.h>
int main() {
    int n, i, j;
    int bt[20], wt[20], tat[20];
    int temp;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    printf("Enter burst time:\n");
    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (bt[i] > bt[j]) {
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }
    wt[0] = 0;
    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }
    printf("\nProcess\tBT\tWT\tTAT\n");
    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }
    return 0;
}
```

Output:

```
rohitsai@MSI:~/OS$ nano SJF_np.c
rohitsai@MSI:~/OS$ gcc SJF_np.c -o SJF_np
rohitsai@MSI:~/OS$ ./SJF_np
Enter number of processes: 3
Enter burst time:
P1: 3
P2: 5
P3: 7

Process BT      WT      TAT
P1      3      0      3
P2      5      3      8
P3      7      8     15
```

3. Priority

```
GNU nano 8.7.1
#include <stdio.h>
int main() {
    int n, i, j;
    int bt[20], priority[20], wt[20], tat[20];
    int temp;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    printf("Enter burst time and priority:\n");
    for (i = 0; i < n; i++) {
        printf("P%d Burst Time: ", i + 1);
        scanf("%d", &bt[i]);

        printf("P%d Priority: ", i + 1);
        scanf("%d", &priority[i]);
    }
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (priority[i] > priority[j]) {
                temp = priority[i];
                priority[i] = priority[j];
                priority[j] = temp;
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }
    wt[0] = 0;
    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }
    printf("\nProcess\tBT\tPriority\tWT\tTAT\n");
    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t\t%d\t%d\n",
            i + 1, bt[i], priority[i], wt[i], tat[i]);
    }
    return 0;
}
```

Output:

```
rohitsai@MSI:~/OS$ nano Priority.c
rohitsai@MSI:~/OS$ gcc Priority.c -o Priority
rohitsai@MSI:~/OS$ ./Priority
Enter number of processes: 4
Enter burst time and priority:
P1 Burst Time: 3
P1 Priority: 7
P2 Burst Time: 5
P2 Priority: 2
P3 Burst Time: 6
P3 Priority: 1
P4 Burst Time: 4
P4 Priority: 8

Process BT      Priority      WT      TAT
P1      6        1          0        6
P2      5        2          6       11
P3      3        7         11       14
P4      4        8         14       18
```

4. ROUND ROBIN

```
rohitsai@MSI: ~/OS
GNU nano 8.7.1
#include <stdio.h>
int main() {
    int n, i;
    int bt[20], rem_bt[20];
    int wt[20], tat[20];
    int quantum;
    int time = 0;
    int done;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    printf("Enter burst time:\n");
    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
        rem_bt[i] = bt[i];
        wt[i] = 0;
    }
    printf("Enter time quantum: ");
    scanf("%d", &quantum);
    while (1) {
        done = 1;
        for (i = 0; i < n; i++) {
            if (rem_bt[i] > 0) {
                done = 0;
                if (rem_bt[i] > quantum) {
                    time = time + quantum;
                    rem_bt[i] = rem_bt[i] - quantum;
                }
                else {
                    time = time + rem_bt[i];
                    wt[i] = time - bt[i];
                    rem_bt[i] = 0;
                }
            }
        }
        if (done == 1)
            break;
    }
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }
    printf("\nProcess\tBT\tWT\tTAT\n");
    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }
    return 0;
}
```

Output:

```
rohitsai@MSI:~/OS$ nano RoundRobin.c
rohitsai@MSI:~/OS$ gcc RoundRobin.c -o RoundRobin
rohitsai@MSI:~/OS$ ./RoundRobin
Enter number of processes: 3
Enter burst time:
P1: 4
P2: 5
P3: 6
Enter time quantum: 2

Process BT      WT      TAT
P1      4        4        8
P2      5        8       13
P3      6        9       15
```